Experiment-16

16. Letter Frequency Attack on Monoalphabetic Substitution Cipher

Aim: To write a C program to perform letter frequency analysis on a monoalphabetic substitution cipher and generate a possible plaintext.

Procedure:

1. Read the ciphertext from the user.
2. Count the frequency of each alphabet.
3. Sort letters according to their frequency.
4. Map frequent ciphertext letters to common English letters.
5. Display the possible plaintext.

Code (q16.c)

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


int main(void) {

  char c[1000], p[1000];

  char eng[] = "ETAOINSHRDLCUMWFGYPBVKJXQZ";

  char map[26];

  int f[26] = {0}, a[26], i, j, top;


  printf("Enter ciphertext: ");

  fgets(c, sizeof(c), stdin);

  c[strcspn(c, "\n")] = '\0';


  printf("Enter number of candidates: ");
```

```
scanf("%d", &top);

if (top < 1) top = 1;
if (top > 26) top = 26;

for (i = 0; c[i]; i++)
   if (isalpha((unsigned char)c[i]))
      f[toupper((unsigned char)c[i]) - 'A']++;

for (i = 0; i < 26; i++) a[i] = i;

for (i = 0; i < 26; i++)
   for (j = i + 1; j < 26; j++)
      if (f[a[j]] > f[a[i]]) {
         int t = a[i]; a[i] = a[j]; a[j] = t;
      }

for (i = 0; i < 26; i++) map[a[i]] = eng[i];

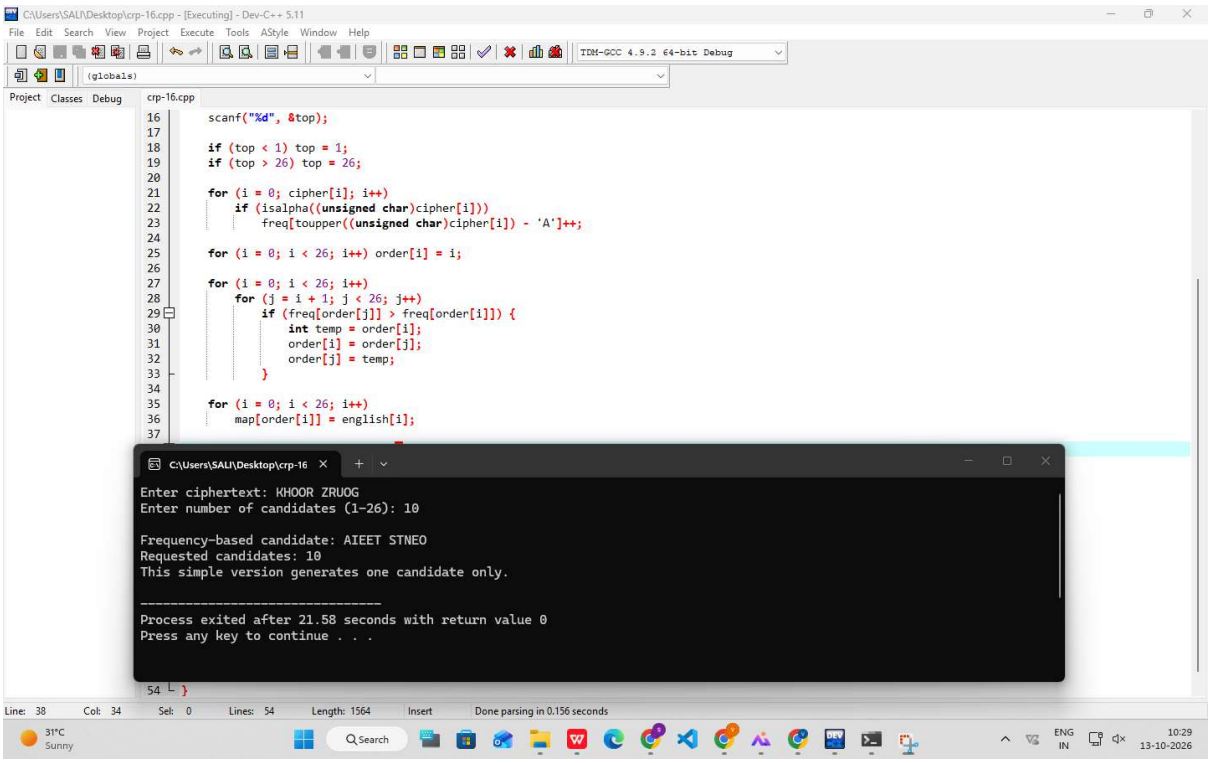
for (i = 0; c[i]; i++) {
   if (isalpha((unsigned char)c[i])) {
      char x = map[toupper((unsigned char)c[i]) - 'A'];
      p[i] = islower((unsigned char)c[i])
         ? (char)tolower((unsigned char)x) : x;
   } else p[i] = c[i];
}
p[i] = '\0';
```

```
    printf("Possible plaintext: %s\n", p);
    printf("Requested top candidates: %d\n", top);
    printf("This basic program produces one candidate.\n");
    return 0;
}
```

Sample Input

```
Enter ciphertext: KHOOR ZRUOG
Enter number of candidates: 10
```

Sample Output



Result:

The program performs letter frequency analysis and generates a possible plaintext. It is a basic approximation and does not guarantee correct decryption.